# Design and Proof of Concept of a Hybrid Cable-Driven Cartesian Robot for High-Density Storage in Hypoxic Environments

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***Abstract*— This paper describes the conceptual design stage and proof of concept of a special-purpose Automated Storage and Retrieval System (AS/RS) developed for the new Document Preservation Center (CPD) of the National Autonomous University of Mexico (UNAM) in Juriquilla, México. This facility will house the historical archives of the National Library and National Hemerotheque of Mexico. Because the main building will feature a hypoxic (reduced oxygen) atmosphere to prevent fires and biological proliferation, a 100% automated system with zero human presence was proposed. This paper analyzes existing commercial solutions and justifies the design of a hybrid system comprising a large-scale cable-driven serial Cartesian robot (55 m long by 15 m high) and autonomous mobile robots. A trade-off analysis between storage density, technical complexity, and material preservation is detailed, resulting in high-density single-aisle architecture. The final installation and commissioning of the system are scheduled for 2027.**

***Keywords*— *Automated Storage, Cartesian Robotics, Cable Robots, Document Preservation, Conceptual Design, AS/RS.***

## I. Introduction

Safeguarding bibliographic and hemerographic heritage represents the conservation of a nation's cultural and historical memory. In this context, the National Autonomous University of Mexico (UNAM) is developing the Document Preservation Center (CPD) in Juriquilla, México. The automation of library environments has been a growing trend, evolving towards architectures integrated with the Internet of Things (IoT) and artificial intelligence for efficient collection management[1], [2]. However, the selection and design of these automated systems must obey rigorous principles that prioritize usability and long-term material preservation over simple logistical throughput [3].

The central problem of this research lies in the specific environmental and handling restrictions of the CPD's core "Building C". To prevent biological proliferation and eliminate fire risks, this 9,600 m³ facility will operate under a hypoxic (reduced oxygen) atmosphere, a condition that prohibits sustained human activity and demands absolute automation. Historically, research in library robotics has focused on interactive service robots [4], teleoperated systems for consultation [5], and semi-autonomous RFID implementations [6] capable of navigating in human-shared spaces [7]. These technological solutions prove ineffective against the need to continuously handle Logistics Units (LU) weighing up to 110 kg within an high-density and extreme environment scheme.

From these limitations, the following research questions arise: How can high storage density be achieved without compromising the delicate handling of the archive? And what is the better kinematic architecture to mobilize heavy loads across 55 meters in depth and 15 meters in height? The state-of-the-art in industrial Automated Storage and Retrieval Systems (AS/RS) demonstrates that while classic travel-time models for two-axis cranes [8], [9] and shuttle-based systems [10] optimize temporal performance, they often sacrifice density and operate under rigid architectures. Contemporary researchers emphasize the need to transition towards flexible design frameworks [11] supported by multi-criteria decision systems [12]. Recent innovations suggest that integrating omnidirectional Automated Guided Vehicles (AGVs) with advanced collision-free navigation algorithms [13] [14] [15] [16] and modular control architectures [17] offers high adaptability.

In parallel, cable-driven robot technology has emerged, featuring manipulators actuated by steel cables [18]. This architecture provides distinct advantages such as low mechanical complexity, extended workspaces, and low inertia, which translates into higher velocities and accelerations [19], [20]. Although mechanical vibration remains a significant challenge for this technology, effective mitigation strategies have been developed, including tuned mass dampers [21] and electromagnetic dampers [22]. These robotic systems offer clear benefits that, according to the current literature, remain unexploited in the field of automated storage systems.

Based on these findings, the hypothesis of this research postulates that a hybrid automated storage system, combining the kinematics of a cable-driven Cartesian robot with the transverse penetration of autonomous mobile robots, will maximize spatial density in a single aisle and satisfy strict conservation requirements. The proposed experimental design is based on a conceptual design methodology and mechatronic synthesis. The installation and commissioning of the CPD's automated storage system are scheduled for the second half of 2027.

## II. Requirements Analysis and Existing Solutions

During the early stages of development, the engineering team established a set of selection criteria merging two traditionally separated fields: mechatronics/logistics engineering and document conservation science. The priorities were defined in the following order:

*1) Long-term preservation of bibliographic and hemerographic material.*
*2) High storage density within the preservation building.*
*3) High reliability and structural technical simplicity.*
*4) Delivery times adequate to researchers' demands.*

A benchmarking study of commercial Automated Storage and Retrieval System (AS/RS) solutions was conducted. While these technologies are highly efficient in commercial distribution centers and e-commerce, they present severe limitations for the CPD. Commercial systems are optimized for high throughput, sacrificing storage density by requiring multiple aisles. Furthermore, they are neither standardized to handle containers with specific conservation requirements nor designed to operate uninterruptedly in hypoxic atmospheres without frequent in-aisle maintenance. This justified the development of a special-purpose machine. In table 1, it shows a comparison of current technologies applied to automatic storage.

TABLE I. Comparison of Automatic Storage Technologies

| System Name | Brief Description | Main Advantages |
|---|---|---|
| **Heavy-Duty Stacker Cranes (Unit-Load)** | Automated cranes on rails to handle pallets or large containers in narrow aisles. | • Load capacity >1,000 kg.<br>• Heights exceeding 40 m.<br>• High industrial durability. |
| **Small-Load Stacker Cranes (Mini-Load)** | Compact cranes designed for fast extraction of individual boxes or trays. | • High picking cycle speed.<br>• Vertical space optimization.<br>• Lower structural energy consumption. |
| **Shuttle Systems** | Fleets of autonomous vehicles per level, supported by vertical lifts at the ends. | • Extremely high throughput via simultaneous operation.<br>• Redundancy against vehicle failure.<br>• High modularity. |
| **Cube Storage (High Density)** | 3D grid of stacked containers without aisles, operated by robots on top. | • Maximum density by eliminating aisles.<br>• Highly scalable.<br>• Low energy consumption. |
| **Vertical Lift Modules (VLM) & Carousels** | Closed structures with a central lift (VLM) or continuously rotating shelves. | • Floor space savings of up to 85%.<br>• Ergonomic goods-to-person operation.<br>• Dust-protected environment. |
| **Autonomous Mobile Robots (AMR)** | Free-navigating vehicles that transport racks or climb 3D structures. | • Total flexibility without heavy fixed infrastructure.<br>• Fast implementation and reconfiguration. |

As evidenced in the benchmarking study, commercial AS/RS technologies inherently force a trade-off between storage density and material safety. For instance, while Cube Storage systems maximize volumetric space, they are incompatible with the delicate handling of 110 kg archival logistics units. On the other hand, Vertical Lift Modules provide excellent protection but completely lack the scalability required for a massive 55-meter-long facility.

This technological gap underscores the inadequacy of repurposing standard e-commerce solutions for historical archives. To overcome these limitations, a paradigm shift in kinematic architecture is required: one that merges the heavy-load stability and reliability of traditional automated cranes with the deep-lane density capabilities of mobile robotics.

## III. Conceptual Design Methodology and Multi-Criteria Evaluation

The system's conceptual design originated from the definition of the "Logistics Unit" (LU), a custom-engineered metal container tailored to the dimensions of bibliographic materials. Its geometry facilitates archival access while shielding the contents from external damage. The mass and spatial footprint of the LU dictate the dynamic, inertial, and structural requirements of the entire mechatronic system. Through iterative design processes guided by the operational and preservation constraints of librarians and conservators, the maximum operational mass was standardized at 110 kg (inclusive of the LU and its documentary payload). To accommodate the diverse geometries of the historical archive, two LU variants were developed: a large-format version (67 cm deep, 51 cm wide, and 52 cm high) and a small-format version (sharing the same footprint but reduced to a 37 cm height). Establishing these foundational parameters was critical for minimizing the cross-sectional area of the structural racks, optimizing the power requirements of the robotic actuators, and ultimately reducing the system's overall energy consumption.

### *A. The Spatial Density and Number of Aisles Problem*

A critical variable in the system's modeling was the inversely proportional relationship between the number of aisles, storage density, and technical complexity. Building C of the CPD has a usable volume defined by its internal dimensions of 11.44 m wide, 55 m long, and up to 15 m high. Figure 1 shows the interior of building C.

To rigorously quantify this constraint, a kinematic and volumetric model was developed to evaluate the structural topology of Building C's 11.44-meter-wide storage area. The analysis demonstrated a severe inverse relationship between the number of aisles ($N_p$) and the facility's total logistics unit (LU) capacity. Deploying a traditional multi-aisle architecture drastically sacrifices spatial efficiency, as each aisle constitutes a dead volume incapable of housing documentary material. For instance, a single-aisle configuration caps the capacity at 28,812 LUs, a standard two-aisle configuration caps the capacity at 26,754 LUs, while scaling up to a five-aisle layout collapses it to merely 14,406 LUs. Furthermore, multiplying the number of aisles inherently inflates the probability of mechanical failures by requiring either multiple active cranes or highly complex aisle-changing mechanisms.

In stark contrast, implementing a single 55-meter-long central aisle maximizes the volumetric frontier. This configuration deploys two massive storage racks on each side, necessitating deep-lane penetration to access the archive. To resolve this challenge without compromising the density of

28,812 LUs, a novel hybrid architecture was synthesized. A large-scale Cartesian robot provides high-speed longitudinal and vertical transport, positioning its end-effector at the target coordinate. Subsequently, an autonomous mobile robot departs from the end-effector to execute a deep transverse penetration into the passive racks. This decoupled kinematic strategy enables ultra-high-density retrieval operations under a First-In, Last-Out (FILO) arrangement, validating its superiority over conventional commercial solutions.



Fig. 1. The interior of Building C is shown, with a person in the background providing a sense of scale. The dimensions are 11.5 m in width, 15 m in height, and 55 m in length (extending away from the viewer). A black prototype rack for logistics units is also visible on the right.

### *B. Multi-Criteria Design Evaluation*

To rigorously justify the choice of this hybrid architecture over purely commercial AS/RS solutions, a weighted decision matrix (Pugh Method) was developed. The evaluation criteria were weighted on a scale of 1 to 5 (5 being the highest priority) based on the demands of the hypoxic environment and heritage preservation. The technologies evaluated against the proposed design include the Industrial Stacker Crane (Unit-Load AS/RS), Multi-level Shuttle Systems, and 3D Lifting Autonomous Mobile Robots (AMR 3D).

TABLE II. WEIGHTED DECISION MATRIX FOR AS/RS SYSTEMS

| Evaluation Criteria | Weight (1-5) | Hybrid Proposal (Cartesian + Mobile) | Stacker Crane (Unit-Load) | Multi-level Shuttles | Mobile Robots (AMR 3D) |
|---|---|---|---|---|---|
| **Storage Density** (Volumetric optimization) | **5** | 5 | 4 | 3 | 4 |
| **Archive Safety & Handling** (Low vibration/impact) | **5** | 5 | 3 | 4 | 3 |
| **Hypoxic Environment Maintenance** (External accessibility) | **5** | 4 | 3 | 2 | 2 |
| **Payload Capacity** (Stable handling of 110 kg) | **4** | 5 | 5 | 3 | 2 |
| **Passive Rack Simplicity** (No electronics in racks) | **4** | 5 | 4 | 2 | 3 |
| **Throughput / Flow Velocity** (Extractions per hour) | **3** | 3 | 2 | 5 | 4 |
| **Total Weighted Score** | -- | **119** | **92** | **80** | **77** |

As observed in Table II, the Hybrid Proposal obtained the superior score due to its alignment with the CPD's critical requirements. While Shuttle systems lead in logistical flow speed, their high mechatronic complexity inside the racks makes them unviable for a reduced oxygen environment where internal maintenance is highly restrictive. Using a single cable-driven Cartesian robot results in a robust column with minimal failure probabilities. By delegating deep collection to the transversal mobile robot, the proposed system ensures the integrity of the Logistics Units and guarantees a spatial density unattainable for standard multi-aisle architectures.

## IV. PROPOSED SYSTEM ARCHITECTURE

The resulting system is discretized into four main subsystems, maintaining operational independence to facilitate maintenance and scalability.

Storage Racks (R). These consist of high-density metal racks located on both sides of the central aisle (55 m long). Their design lacks internal moving mechanisms, ensuring passive long-term reliability and minimizing failure points. Inside these racks, up to seven logistics units are stored in depth. When fully loaded, the warehouse is expected to hold 28,812 logistics units.

Cartesian Robot (CR). It is the backbone of the system. It consists of a large-scale cable-driven serial manipulator with three degrees of freedom: two linear displacements and a 180° rotational movement at the end effector, allowing it to align the mobile robot with both the left and right racks. The CR moves along rails at the bottom and top of Building C's aisle.

- X-Axis (Horizontal displacement): A translation carriage runs at floor level along the 55 m of the building, reaching a maximum speed of 2 m/s with accelerations of 0.15 m/s$^2$. It is driven by a steel cable circuit attached

to a servo-motor reel located externally in the engine room.

- Y-Axis (Vertical displacement): A 15-meter-high structural column along which the end effector ascends, reaching a maximum speed of 2 m/s with accelerations of 0.56 m/s$^2$. It is driven by a steel cable attached to a servo-motor reel externally located in the engine room.

Cable-Driven Transmission. The horizontal linear displacement of the Cartesian robot is performed by a closed-loop stainless steel cable circuit. The circuit starts at the wheeled base of the Cartesian robot, moves to the front where there is a return pulley, and passes to the back of Building C into the engine room. There, the cable wraps around a special reel driven by a servomotor to generate the horizontal movements. The vertical linear displacement is similar, utilizing an open-loop steel cable attached at one end to the end effector and at the other to a reel in the engine room.

Mobile Robot (MR) and End Effector (EE). The CR's End Effector includes a 180° rotary table, powered by batteries and wireless communication, which carries the autonomous vehicle, denominated Mobile Robot (MR). Upon reaching the desired (x, y) coordinate, the MR departs from the EE, venturing deep into the SR to lift and extract the Logistics Unit. The MR/LU assembly returns to the EE, which finally delivers it to the transition room. This zone consists of hermetic airlocks (pass-boxes), after which users such as librarians or conservators can safely access the materials. Figure 2 shows a diagram with the main elements involved in the system.

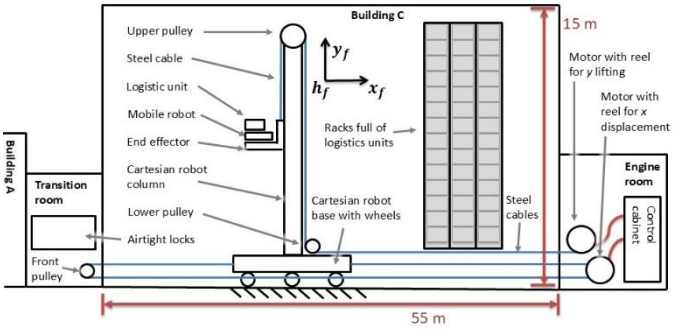



Fig. 2. *Diagram of the elements involved in the automated storage system.*

This hybrid configuration offers substantial advantages:

- Maintains high storage density by requiring a single aisle.
- Improves material safety by reducing failure probabilities through a single, robust Cartesian robot.
- Allows variable delivery speeds using multiple mobile robots in parallel, adapting to request peaks without modifying the main structure.
- By implementing steel cables, the need to incorporate heavy motors directly on the CR is eliminated, which improves accelerations and reduces energy consumption.

Figure 3 shows the Computer-Aided Design (CAD) of the components.

## V. Preliminary Results (Proof of Concept)

To validate the technical and kinematic viability of the proposed hybrid architecture, a physical and functional Proof of Concept (PoC) prototype was developed with scaled-down travel distances. The experimental setup (see Figure 4) integrated all critical subsystems described in the conceptual design: rack sections configured with a single central aisle, the Cartesian Robot (CR), the End Effector (EE) equipped with its rotary table, the Mobile Robot (MR), and physical samples of the Logistics Units (LU).

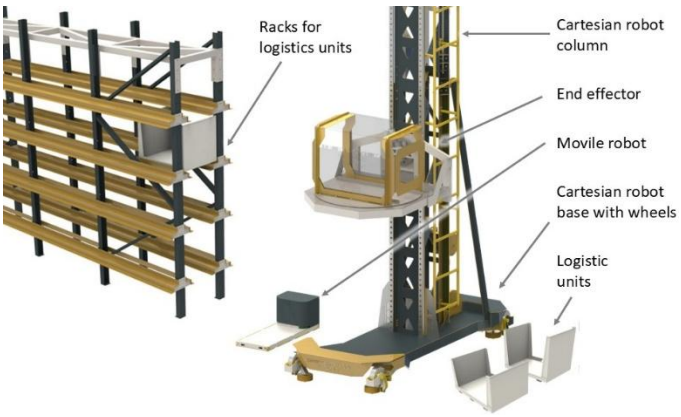



Fig. 3. *CAD model of the main elements of the automated storage system.*

Unlike the final design projected for the CPD, this prototype limited its horizontal travel stroke to 2 m (equivalent to housing slightly more than three columns of logistics units) and its vertical elevation to 2 m. These stroke reductions were established to operate safely within a controlled laboratory environment. The experimental tests focused on evaluating the mechatronic integration and the structural stability of the robotic interfaces.

The coupling and decoupling kinematics were successfully validated. Specifically, the fluid transition of the Mobile Robot from the End Effector's rotary table into the rails of the Storage Racks was achieved with a mechanical tolerance of +/- 5 mm, completing the safe clamping and extraction sequence of the real Logistics Unit efficiently. These preliminary results confirm that the strategy of separating the massive translation axes (Cartesian robot) from the transverse penetration axes (mobile robot) is mechanically robust. It proves that the design concept meets the strict stability requirements demanded for handling historical archives. Fig. 4 shows the built and tested prototype. Figure 5 displays a photograph of the lower section of the full-scale Cartesian robot, fully manufactured and nearly ready for installation and commissioning.




Fig. 4. *Photograph of the PoC prototype of the automated storage system, integrating the CR, EE, MR, and Racks.*

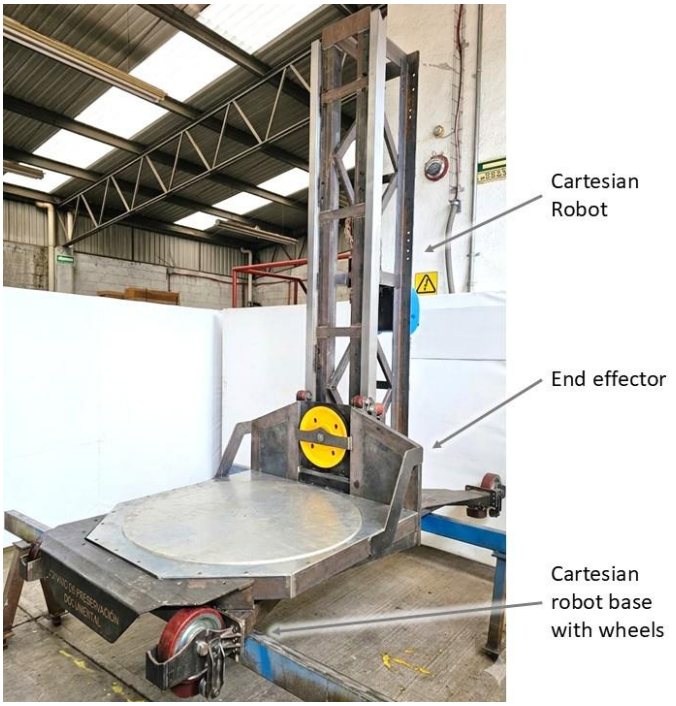



Fig. 5. *Lower section of the manufactured full-scale Cartesian robot, nearly ready for installation and operation.*

## I. Conclusions

The development of the automated storage system for the Document Preservation Center (CPD) of UNAM demonstrates how extreme conservation demands can drive innovative mechatronic solutions. By abandoning the paradigm of commercial AS/RS systems exclusively oriented towards continuous flow and opting for a special-purpose architecture, this research contributes to the state-of-the-art by proving the technical viability of hybrid robotic systems in restrictive environments. The integration of a cable-driven Cartesian Robot with Mobile Robots allowed the consolidation of a single-aisle design that maximizes storage density and reduces the complexity of the passive infrastructure (the storage racks). This responds affirmatively to the initial hypothesis regarding the feasibility of balancing high density, structural safety, and delicate material handling.

The results of this design phase gain particular relevance when contrasted with prior knowledge. Unlike traditional service robotics platforms implemented in libraries, which lack the industrial load capacity necessary to handle 110 kg logistics units, the proposed solution scales heavy industry principles without sacrificing bibliothecological standards. Furthermore, kinematic analyses validate that by decoupling the transversal transport (mobile robot) from the main lifting transport, the mechanical redundancies and high energy consumption typical of commercial multi-level shuttle systems are avoided. Thus, the implementation of this system during 2027 marks a technological milestone in engineering applied to conservation in Latin America, bridging the gap between industrial logistics and heritage preservation in hypoxic atmospheres.

From a practical application perspective, the design principles and centralized control architecture developed in this project are easily extrapolated to other industrial sectors. Any storage environment demanding high density and absolute restriction of human presence, such as hazardous materials storage, extreme cold rooms, or biotechnology repositories, can benefit from this separation between passive structures and mobile effectors transported by cable-driven cartesian robots.

The following phases of the research will focus on the empirical validation of the system during its installation and commissioning in the second half of 2027. Next steps include tuning the dynamic control laws to mitigate vibrations and stretching in the Cartesian robot's cable actuators, validating theoretical elongation models. Likewise, experimental evaluation of collision-free routing algorithms for vehicles in the Transition Zone (TZ) under real operating conditions will be conducted. Finally, a comparative study between physically measured cycle times and analytical travel time models is planned, aiming to optimize the software interface and library request scheduling.

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